

Lecture2

September 9, 2026

1 Lists

```
[20]: my_list = ['item 0', 1, 2.0, 'end']  
      # You can have a bunch of stuff in it, different data types  
  
      for i in my_list:  
          print(i)
```

```
item 0  
1  
2.0  
end
```

```
[22]: # Accessing items in the list. Using index  
      first_item = my_list[0]  
      third_item = my_list[2]  
  
      last_item = my_list[-1]  
  
      print(first_item)  
      print(third_item)  
      print(last_item)
```

```
item 0  
2.0  
end
```

```
[28]: # Slicing a list. Access a chunk of items. Specify a range of indices  
      # The last one is not included. It is exclusive. Inclusive to exclusive.  
      first_slice = my_list[1:3]  
  
      # Get to the end  
      to_end = my_list[2:]  
  
      # From beginning  
      from_start = my_list[:3]  
  
      print(first_slice)
```

```
print(to_end)
print(from_start)
```

```
[1, 2.0]
[2.0, 'end']
['item 0', 1, 2.0]
```

```
[30]: # Methods
      # Length
      print(len(my_list))
```

```
4
```

```
[46]: # Append, add stuff
      my_list.append('true end')

      print(my_list)
```

```
['item 0', 1, 2.0, 'second last', 'true end', 'true end', 'true end', 'true
end', 'true end']
```

```
[48]: my_list[3] = 'second last'
      print(my_list)
```

```
['item 0', 1, 2.0, 'second last', 'true end', 'true end', 'true end', 'true
end', 'true end']
```

```
[53]: # Tuple, immutable data type
      # use it if you want to not change your dimensions
      my_tuple = (1, 2, 3)
```

```
[57]: # Range function
      # Default increment is 1
      # Range is exclusive
      list_by_range = list(range(-2,4))
      print(list_by_range)

      # Third number is the increment
      list_by_range = list(range(4, -5, -3))
      print(list_by_range)
```

```
[-2, -1, 0, 1, 2, 3]
[4, 1, -2]
```

```
[61]: # For loop
      # i is the loop variable
      for i in my_list:
          print(i)
```

```
item 0
1
2.0
second last
true end
true end
true end
true end
true end
```

```
[71]: # Example
cubes = []

for num in range(1, 11):
    cube = num**3
    cubes.append(cube)

print(cubes)
```

```
[1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
```

```
[73]: s = 0
for k in range(1, 101):
    s += k
print(s)
```

```
5050
```

```
[ ]:
```